Project Documentation Guidelines

Egypt Digital Pioneers (DEPI)
Ministry of Communications and Information Technology

Project Deadlines

Item	Student Deadline	Graduate Deadline
Project Planning & Management	2/20/2026	1/27/2026
Literature Review	4/20/2026	3/1/2026
Requirements Gathering	4/20/2026	3/1/2026
System Analysis & Design	5/1/2026	4/3/2026
Implementation (Source Code & Execution)	7/10/2026	5/17/2026
Final Presentation & Testing & Reports	7/17/2026	5/24/2026

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Submission Date: April 2026

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1. Project Planning & Management

Project Proposal:

This project develops an interactive Power BI dashboard for Maven Fuzzy Factory to transform Raw E-commerce data into business insights.

Project Scope:

The scope of this project is to develop an interactive **Business Intelligence dashboard using Power BI** for Maven Fuzzy Factory. The project focuses on analyzing historical business data related to sales, products, customers, website traffic, and marketing campaigns in order to generate insights, monitor KPIs, and support data-driven decision-making.

In Scope

The following activities and features are included in the project:

- Importing datasets from Excel / CSV / database sources into Power BI.
- Cleaning and transforming raw data using **Power Query**.
- Building a structured data model with relationships between tables.
- Creating calculated measures and KPIs using **DAX**.
- Designing interactive dashboard pages for:
    - Sales Performance
    - Product Analysis
    - Customer Behavior
    - Marketing Campaign Performance
    - Website Traffic Analysis
- Using slicers, filters, drill-through, and date analysis.
- Visualizing trends through charts, cards, maps, and tables.
- Generating business insights and recommendations based on dashboard findings.
- Publishing the final Power BI report for presentation purposes.

Out of Scope

The following items are outside the current project scope:

- Building transactional systems for sales or inventory operations.
- Developing a website or mobile application.
- Real-time data streaming dashboards.
- Advanced machine learning or AI models.
- Automatic data entry systems.
- ERP / CRM system integration.
- User role security implementation at enterprise level.
- Custom coding outside standard Power BI tools.

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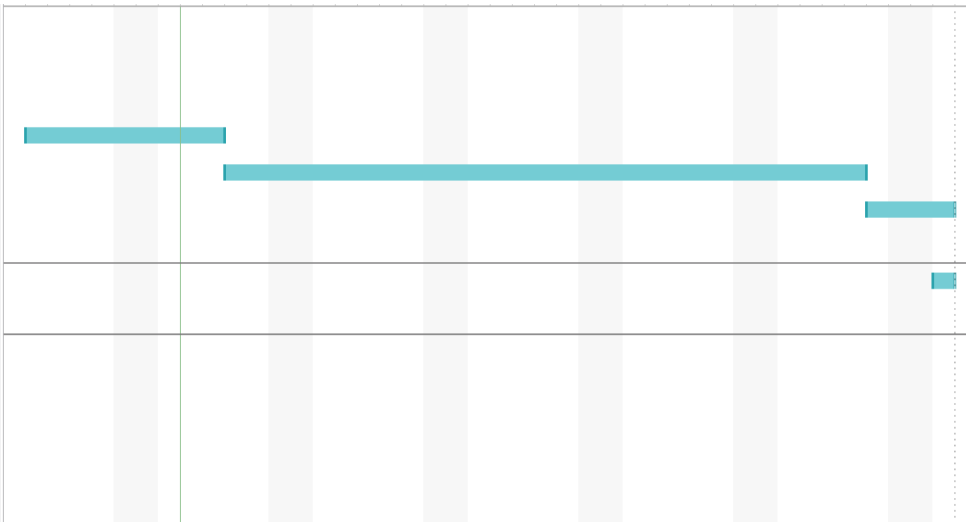
Project Plan:

		Tue 27/01/26	
Project Planning & Management	19 days	Tue 27/01/26	Fri 20/02/26
Literature Review	13 days	Wed 25/03/26	Fri 10/04/26
Requirements Gathering	7 days	Tue 21/04/26	Wed 29/04/26
System Analysis & Design	21 days	Thu 30/04/26	Thu 28/05/26
Implementation (Source Code & Execution)	2 days	Fri 29/05/26	Mon 01/06/26
Final Presentation & Testing & Reports	1 day	Mon 01/06/26	Mon 01/06/26



2.Project Milestones:

Milestone	Duration	Description
Project Planning	Week 1	Define KPIs, business questions, dashboard scope
Data Collection	Week 2	Gather sales, products, campaigns, customer data
Data Cleaning	Week 3	Use Power Query to clean and transform data
Data Modelling	Week 4	Build relationships between tables
KPI Development	Week 5	Create measures using DAX
Dashboard Design	Week 6–8	Build interactive report pages
Testing	Week 9	Validate numbers and visuals
Final Documentation	Week 10	Final report + explanation
Presentation	Week 11	Demo and project presentation

3. Deliverables:

Phase	Deliverable
Planning	KPI List + Project Proposal
Data Prep	Clean Dataset
Modelling	Star Schema Model
Development	Power BI Dashboard (.pbix)
KPIs	DAX Measures
Final Output	Report + Presentation

4.Resource allocation:

Role	Responsibility	Allocation
Team Leader	Manage timeline and report	20%
Data Analyst	Cleaning + KPI logic	30%
Power BI Developer	Dashboard visuals	35%
Presenter / Tester	Testing + Presentation	15%

Task Assignment Roles:

Name	Role
Omar Mohamed Farook (Team Leader)	**Data Modeling & Dashboard Design**
Samar Hassan Abbas	**Data Cleaning & Dashboard Design**
Samira Gamal Mohamed	**Data Exploration & Dashboard Design**
Amr abdel hakeem Mohamed	**Data Visualization & Dashboard Design**
Rania Adel Sakr	**Reporting & Dashboard Design**

Risk Assessment:
• Missing data → cleaning process
• Duplicate records → remove duplicates
• Slow performance → optimized model
• KPI mismatch → formula validation

KPIs (Key Performance Indicators):

KPIs & Insights

- **Total Revenue:** Measures the total income generated from all sales and indicates overall business performance.
- **Total Orders:** Shows the number of completed purchases and reflects customer demand.
- **Conversion Rate (CVR %):** Indicates the percentage of visitors who placed orders, measuring sales effectiveness.
- **Average Order Value:** Reveals the average amount spent per transaction and helps evaluate customer spending behavior.

- **Profit Margin %:** Shows how much profit is earned from revenue after costs, reflecting profitability.
- **Total Items Sold:** Measures the total quantity of products sold and highlights product demand.
- **Avg Items per Order:** Indicates how many items customers buy per order, showing basket size trends.
- **Total Pageviews:** Tracks the number of pages viewed, reflecting customer interest and browsing activity.
- **Total Sessions:** Measures total website visits and helps evaluate traffic performance.
- **Refund % of Revenue:** Shows the share of revenue lost through refunds, indicating return issues or satisfaction levels.

2. Literature Review

Feedback & Evaluation

The project shows a very good effort in using Power BI to analyze business data and present it in a professional dashboard format. The KPIs selected are relevant and help in understanding sales performance, customer behavior, and marketing results clearly. The dashboard is interactive, organized, and easy to use, which reflects a good understanding of Business Intelligence concepts. Overall, the project is well done and achieves its main objective successfully.

Suggested Improvements

- Add more advanced DAX calculations for deeper analysis.
- Include forecasting features to predict future sales trends.
- Improve dashboard design consistency with a stronger visual theme.
- Add more drill-through pages for detailed product insights.
- Use live data connections instead of static datasets.
- Optimize report performance and loading speed.
- Add written insights or recommendations on each dashboard page.
- Consider implementing user access levels in future versions.

3. Requirements Gathering

Stakeholders:
• Management
• Sales Team
• Marketing Team
• Analysts

User Stories:

• **User Stories & Use Cases**

These scenarios explain how users interact with the Maven Fuzzy Factory online store and how their behavior is reflected in the Power BI dashboard.

User Story 1 – Customer Arrives from Advertisement and Makes a Purchase

A customer sees a marketing advertisement on social media and clicks the link to visit the Maven Fuzzy Factory website. After entering the site, the customer browses the homepage, scrolls through featured toy products, and opens a product page to view details. The customer likes the product, adds it to the cart, completes the purchase, and exits the website.

Business Value: This journey increases sessions, pageviews, conversion rate, orders, and revenue while showing successful campaign performance.

User Story 2 – Customer Visits but Does Not Purchase

A customer clicks on an online advertisement and enters the website. The customer browses several pages, checks product prices, and views one product page. After reviewing the product, the customer leaves the website without adding anything to the cart or completing a purchase.

Business Value: This journey increases sessions and pageviews but lowers conversion rate, indicating a need to improve product appeal or pricing strategy.

User Story 3 – Returning Customer Purchases Multiple Items

A previous customer returns directly to the website after a positive earlier experience. The customer quickly selects two or more toy products, adds them to the cart, and completes the checkout process.

Business Value: This journey improves repeat customer rate, total items sold, average order value, and customer loyalty metrics.

User Story 4 – Customer Requests Refund

A customer purchases a toy product but later requests a refund due to dissatisfaction or delivery issues. The refund is processed successfully.

Business Value: This journey impacts refund percentage of revenue and highlights areas for quality or service improvement.

User Story 5 – High Traffic Campaign Performance

A marketing campaign attracts many visitors to the website, generating a large number of sessions and pageviews. However, only a small number of customers complete purchases.

Business Value: This shows strong campaign reach but weak conversion performance, suggesting the need for better targeting or website optimization.

Functional Requirements:

• Display KPIs
• Filtering by date/product/source
• Revenue trends
• Product comparison

Non-Functional Requirements:

• Fast loading
• Reliable calculations
• Easy navigation

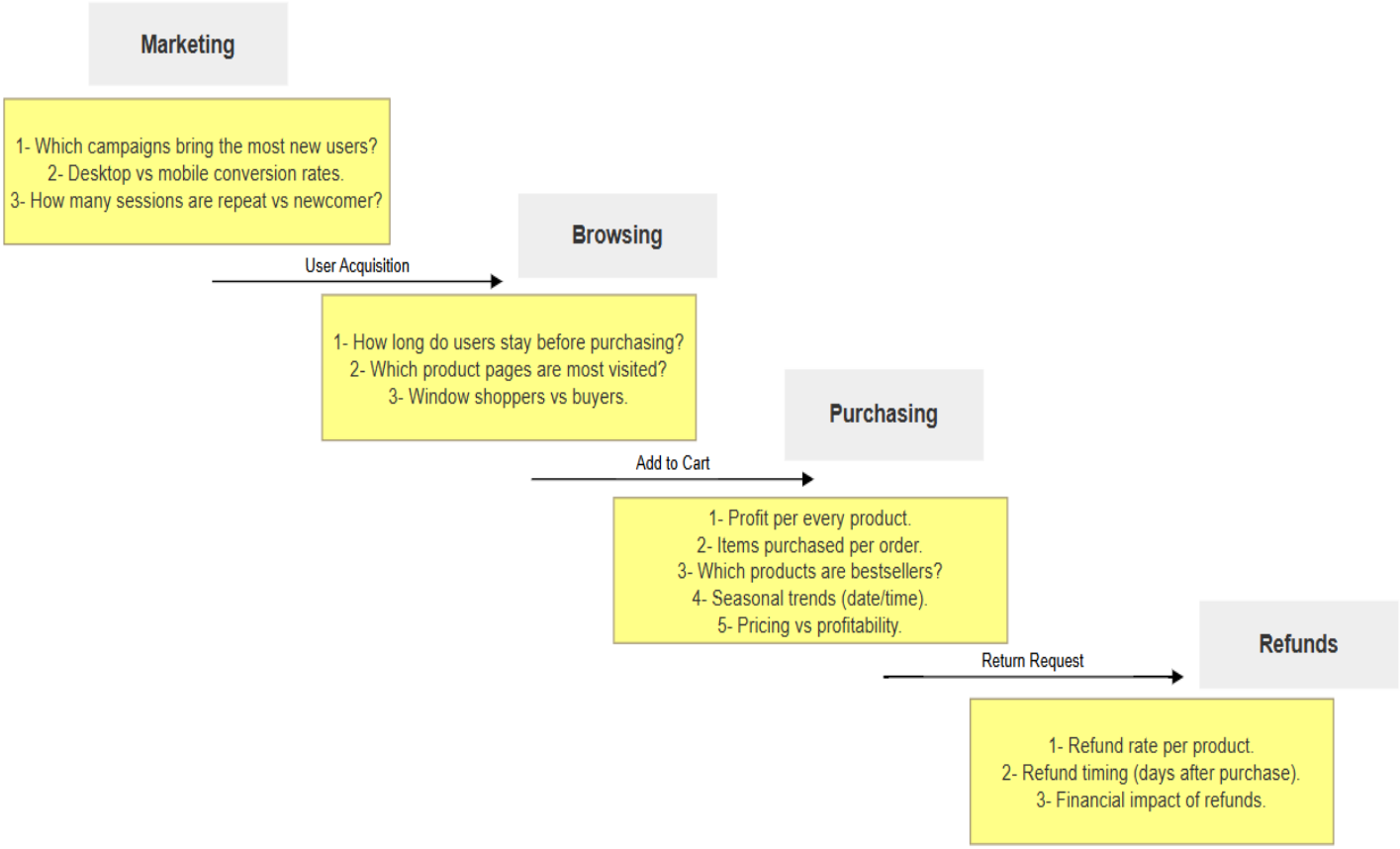
4. System Analysis & Design

Problem Statement:
The company lacks a centralized analytics platform for decision-making.

Data Sources:
website_sessions, orders, order_items, products, traffic_sources

Data Exploration :

business flows **Marketing → Browsing → Purchasing → Refunds.**



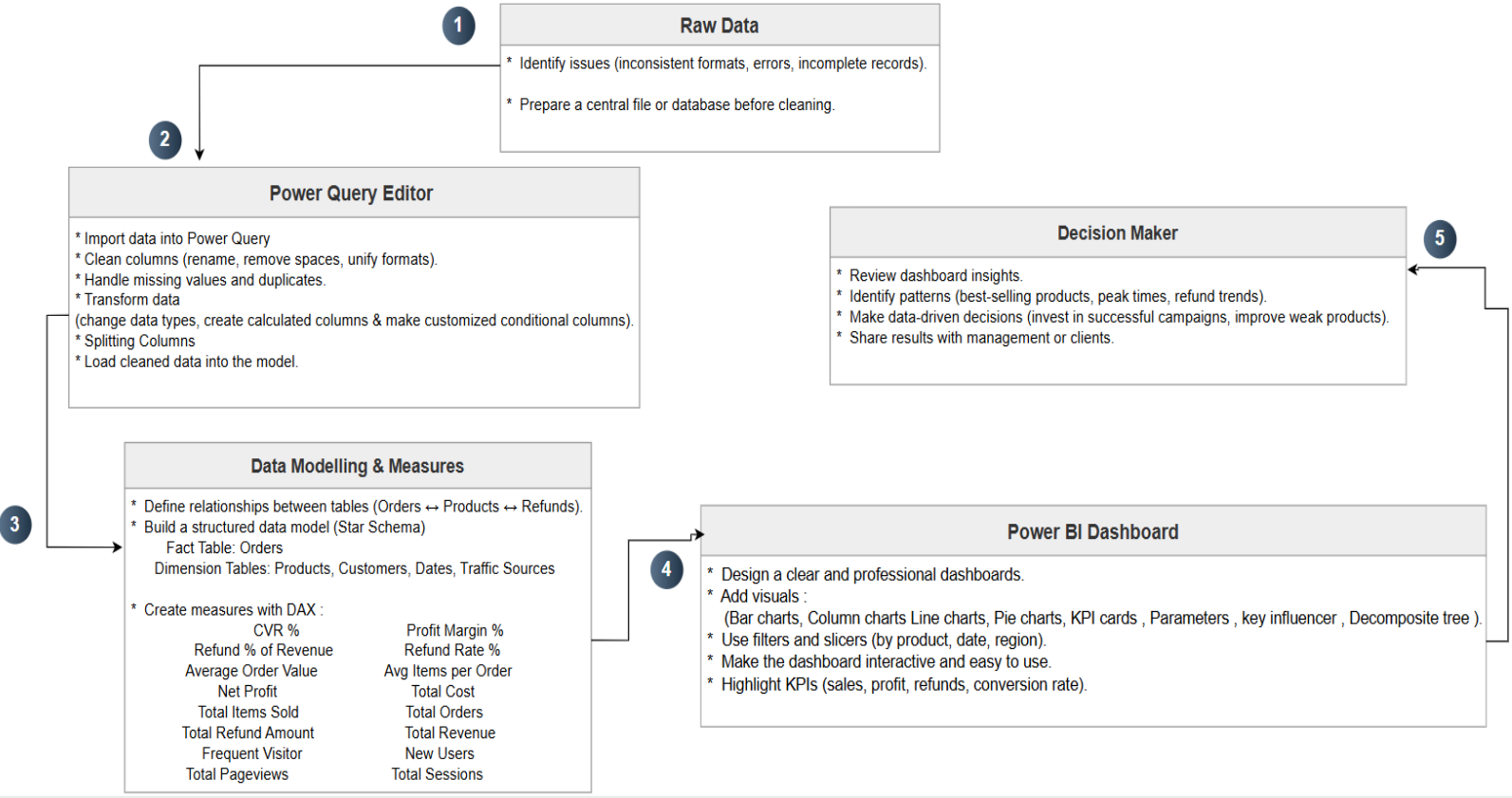
Data Cleaning:

- Handle all Missing values (null)
- Remove duplicates
- Splitting Columns
- Correct data type issues
- Rename columns
- Make conditional Columns (Customized)

Star Schema Model:

Fact Table: Orders

Dimension Tables: Products, Customers, Dates, Traffic Sources



Data Flow:

Raw Data → Power Query → Data Model → Power BI Dashboard → Decision Maker

Dashboard Pages:
1. Executive Summary
2. Sales Performance
3. Marketing Analytics
4. Customer Insights

5. Dashboard KPIs & Insights

KPIs:

- Total Revenue
- Total Orders
- Conversion Rate (CVR %)
- Average Order Value
- Profit Margin %
- Total Items Sold
- Avg Items per Order
- Total Pageviews
- Total Sessions
- Refund % of Revenue

Expected Insights:
• Best-selling products
• Highest revenue periods
• Best traffic channels
• Customer purchase patterns

6. Conclusion

This dashboard converts Maven Fuzzy Factory data into clear visual insights that improve strategic decision-making and operational performance.

7. Future Enhancements

• AI forecasting
• Real-time refresh
• Mobile dashboard
• Automated alerts